Module 1: Fundamentals, OOP Concepts & JVM Architecture

1. **Explain the Object-Oriented Programming Paradigm, Features, and 4 Pillars:** (7 Times - PYQ+QB)
2. **Compare and contrast Structural Programming (C) and Object-Oriented Programming (Java):** (3 Times - PYQ+QB)
3. **Compare Java and C++ Programming Languages:** (2 Times - QB)
4. **Explain the detailed working of JVM (Java Virtual Machine) with a diagram:** (2 Times - PYQ+QB)
5. **Explain how Java is platform independent:** (1 Time - PYQ)
6. **Describe the concept of type casting and type conversion in Java:** (1 Time - PYQ)
7. **Write short note on JDK:** (1 Time - PYQ)

Module 2: Classes, Objects, Methods & Constructors

1. **Explain the different Access Specifiers in Java (`public`, `private`, `protected`, `default`) with examples:** (5 Times - PYQ+QB)
2. **What is Constructor Overloading? Write a program to calculate Area using Constructor Overloading:** (4 Times - PYQ+QB)
3. **Elaborate Argument Passing Mechanism / Object as Argument and returning Object:** (4 Times - PYQ+QB)
4. **What are different types of variables used in Java (Local, Instance, Class/Static)?:** (2 Times - PYQ+QB)
5. **Explain static data members and methods:** (2 Times - PYQ+QB)
6. **Write a JAVA program to create class `Rectangle` and calculate area and perimeter using constructor:** (3 Times - PYQ+QB)
7. **Explain the concept of 'class' and 'object':** (1 Time - PYQ)
8. **Write a program in Java to find nCr and nPr:** (1 Time - PYQ)
9. **WAP to calculate and display the sum of first n natural numbers:** (1 Time - PYQ)

Module 3: Arrays, Matrices & Strings

1. **Elaborate the concept of the `String` class with its methods vs `StringBuffer` class:** (4 Times - PYQ+QB)
2. **WAP to calculate sum and average of array elements in Java with user input:** (3 Times - PYQ)
3. **Write a program to print Fibonacci series using recursion:** (3 Times - PYQ)
4. **Write a program to check whether a string is a Palindrome or reverse a string:** (3 Times - PYQ+QB)

5. **Write a JAVA program to accept m×nm×n matrix and display it in natural form:** (3 Times - PYQ+QB)
6. **State the difference between `String` and `StringBuffer` class in Java:** (2 Times - PYQ+QB)
7. **WAP to find Transpose of a matrix / Matrix Addition:** (2 Times - PYQ+QB)
8. **Write a program to print the Factorial of a number:** (3 Times - PYQ+QB)
9. **Describe any five methods of String Buffer class:** (1 Time - PYQ)
10. **Explain String and StringBuffer classes with an example:** (1 Time - PYQ)
11. **Write Java code for Matrix Multiplication:** (1 Time - QB)

Module 4: Inheritance, Abstract Classes, Interfaces & Packages

1. **Discuss the importance of Inheritance. List all types of inheritance and explain them in detail:** (5 Times - PYQ+QB)
2. **Consider a university where students get fixed grace marks (Sports & Academics): WAP using Interface and Base Class:** (4 Times - PYQ+QB)
3. **WAP to calculate volume of a sphere using Multilevel Inheritance:** (4 Times - PYQ+QB)
4. **Explain Method Overloading vs Method Overriding in detail:** (3 Times - PYQ+QB)
5. **Differentiate between Abstract Class and Interface:** (2 Times - PYQ+QB)
6. **Describe the use of `super` keyword in Java (Constructor, Method, Variable) with an example:** (2 Times - PYQ+QB)
7. **Describe the use of `final` keyword in Java (Variable, Method, Class) with an example:** (2 Times - PYQ+QB)
8. **What is a Package? Explain types of packages and how to create/execute user-defined packages:** (2 Times - PYQ+QB)
9. **Explain multiple inheritance with suitable example:** (1 Time - PYQ)
10. **What is single inheritance with an example?:** (1 Time - PYQ)
11. **Write a program to demonstrate use of interface in Java:** (1 Time - PYQ)
12. **Write a Java program to create an abstract class Shape and derive sphere, cone and cylinder:** (1 Time - PYQ)
13. **Write a program in Java which has an abstract class solid and implements Cylinder and Sphere:** (1 Time - PYQ)

Module 5: Exception Handling & Multithreading

1. **WAP to accept 2 integers from the user, perform division, and demonstrate handling of Exception (divide by zero) using try-catch block:** (6 Times - PYQ+QB)
2. **Illustrate the Life Cycle of Thread in Java with a neat diagram:** (4 Times - PYQ+QB)

3. **How are Threads created? Explain extending `Thread` class vs implementing `Runnable` interface:** (4 Times - PYQ+QB)
4. **WAP for Multithreading (e.g., One thread displays odd numbers, the other displays even numbers):** (4 Times - PYQ+QB)
5. **Explain Exception handling in Java (Discuss try and catch keywords):** (3 Times - PYQ+QB)
6. **Explain the multiple try and catch / Write a program for multiple catch statements:** (3 Times - PYQ+QB)
7. **Explain Custom / User-Defined Exceptions with a Java program:** (2 Times - PYQ+QB)
8. **What is the significance of the finally block in Java exception handling?:** (1 Time - PYQ)
9. **Difference between Errors and Exceptions in Java:** (1 Time - QB)
10. **Explain methods under Thread class (e.g., `start()`, `run()`, `sleep()`, `join()`, `yield()`):** (1 Time - QB)
11. **WAP to demonstrate the making of a thread to print numbers from 1 to 10:** (1 Time - PYQ)

Module 6: GUI Programming (AWT & Swing)

1. **Differentiate between AWT and Swing package of Java:** (6 Times - PYQ+QB)
2. **List and explain various Layout Managers in JAVA GUI / Swing package:** (5 Times - PYQ+QB)
3. **Write a graphics JAVA program to display "All the best" using frame:** (2 Times - PYQ)
4. **Write a java program to draw a smiling Face using Graphics Class Methods:** (1 Time - PYQ)
5. **Write a Java Swing program to create a simple calculator GUI:** (1 Time - PYQ)
6. **Explain Java Swing Class Hierarchy with a block diagram:** (1 Time - QB)
7. **Write a Short Note on Java Swing package (`javax.swing`):** (1 Time - QB)